

Multiple Choice

1. **Answer:** C

Options: A) 1; B) -1; C) i; D) -i

Note: The inverse of -i in the multiplicative group $\{1, -1, i, -i\}$ is i because the product of -i and i equals 1, which is the identity element in the group.

2. **Answer:** A

Options: A) 1; B) 2; C) 5; D) 11

Note: The order of the factor group $(\mathbb{Z}_{11} \times \mathbb{Z}_{15})/(\langle 1, 1 \rangle)$ is 1 because the subgroup $\langle 1, 1 \rangle$ is equal to the entire group $\mathbb{Z}_{11} \times \mathbb{Z}_{15}$, indicating that the factor group contains only the identity element.

3. **Answer:** C

Options: A) 1; B) 2; C) 3; D) 4

Note: In a convex polygon, the sum of the interior angles is determined by the formula $(n-2) \times 180^\circ$, and since each acute angle is less than 90° , the maximum number of acute angles in a convex 10-gon is constrained to 3 because the remaining angles must compensate to maintain the total angle sum of 1440° .

4. **Answer:** C

Options: A) 25; B) 5; C) 6; D) 30

Note: The cyclic subgroup generated by 25 in $\mathbb{Z}_{30}$ has 6 elements because the order of the subgroup is determined by the greatest common divisor of 25 and 30, which is 5, and the number of elements in the subgroup is given by $\mathbb{Z}_{30}$ divided by this order, resulting in $30/5 = 6$.

5. **Answer:** B

Options: A) $x^2/9$; B) $x^2/8$; C) $x^2/4$; D) x^2

Note: The correct answer is derived from applying the formula for the area of a rectangle, where the maximum area is achieved when the length is twice the width, leading to the relationship $A = (x/2)(x/4) = x^2/8$.

True / False

6. **Answer:** True

Options: A) True; B) False

Note: A function can have a local maximum at a point where its derivative does not exist or is not equal to zero, such as at a cusp or corner point, where the function may still attain a higher value compared to its immediate surroundings.

7. **Answer:** True

Options: A) True; B) False

Note: Every free abelian group is torsion-free because it can be generated by a set of linearly independent elements, which allows it to be expressed as a direct sum of copies of integers, where each element has infinite order.

8. **Answer:** False

Options: A) True; B) False

Note: The equation $z^2 = |z|^2$ represents a set of points that includes both real and imaginary parts, leading to a circle in the complex plane.

8. **Answer:** False

Options: A) True; B) False

Note: The probability that the player who makes the first toss wins the game is not $1/4$ because the winning probability depends on the specific rules of the game and the number of possible outcomes, which is often greater than $1/4$.

9. **Answer:** False

Options: A) True; B) False

Note: A finite integral domain can have prime characteristic, as demonstrated by finite fields, which are integral domains with a prime number of elements.

Long Form Response

11. **Answer:** If the composition of functions $g \circ f$ is injective, it can be concluded that the function f must also be injective. This is because, for $g \circ f$ to be one-to-one, each distinct input x_1 and x_2 in the domain of f must lead to distinct outputs in the codomain of g . If f were not injective, there would exist at least two distinct inputs x_1 and x_2 such that $f(x_1) = f(x_2)$, which would cause $g(f(x_1))$ to equal $g(f(x_2))$, contradicting the injectivity of $g \circ f$. Therefore, the injectivity of the composition implies that f itself must be a one-to-one function.

Note: correct because the injectivity of the composition $g \circ f$ necessitates that distinct inputs in the domain of f map to distinct outputs, meaning that if f were not injective, it would lead to equal outputs in $g \circ f$, contradicting the assumption of injectivity.

12. **Answer:** The intersection of two 4-dimensional subspaces V and W within a 7-dimensional vector space X can have dimensions ranging from 0 to 4. However, a dimension of 0 for the intersection is not feasible because it would imply that V and W share no common vectors except the zero vector. Given that both subspaces are 4-dimensional, the maximum possible dimension of their intersection, according to the formula for the dimension of the intersection of two subspaces, is $4 - (\dim V + \dim W - \dim X) = 4 - (4 + 4 - 7) = 1$. Therefore, the intersection must have a dimension of at least 1, meaning they must share some non-zero vector, ruling out the possibility of a dimension of 0.

Note: The reasoning is correct because the dimension of the intersection of two subspaces V and W is constrained by the dimensions of the subspaces and the ambient

space, and a dimension of 0 would contradict the existence of common vectors in the context of their 4-dimensionality within a 7-dimensional space.

End of Answer Key